

BHAVISH GOYAL

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Education

University of Waterloo

Sep 2023 – April 2028

Bachelor's of Computer Science, Specialization : Artificial Intelligence

Waterloo, Canada

- Awarded \$22,000 International Excellence Scholarship for academic distinction.
- Undergraduate Representative at Mathematics Endowment Fund

Work Experience

Communitech Corporation

May 2026 – Present

Software Developer - AI Adoption (Current Co-op)

Waterloo, Canada

Information Systems & Technology (IST), University of Waterloo

Jan 2026 – Apr 2026

AI Developer

Waterloo, Canada

- Built and productionized enterprise AI services using **C#/.NET 8, Azure Functions, Cosmos DB, and Azure OpenAI**, delivering scalable API and retrieval workflows that handled **900+ inputs per request**.
- Enhanced answer relevance by improving the **AI feedback loop** and testing **Cohere reranking, BM25, and contextual retrieval**, boosting retrieval quality by **16%** and reducing hallucinations by **25%**.
- Improved system reliability by implementing **retry logic, structured logging, queue-based workflows, Azure Key Vault, and dependency health checks**, cutting transient failures by **28%** across **3+ cloud integrations**.

Technovia Services

May 2025 – Aug 2025

Software Engineer Intern

Noida, India

- Built **Python**-based data pipelines and **ML evaluation** scripts validating **15K model predictions** via REST APIs, improving **dataset accuracy by 12%** and streamlining analytics for internal testing.
- Implemented **React + TypeScript** dashboards to visualize training accuracy, inference latency, and feature drift, integrating with **backend APIs** for real-time performance tracking and anomaly detection.
- Automated preprocessing and endpoint testing using **Pytest/Postman**, catching 30+ API inconsistencies early and improving deployment reliability by **15%** across development cycles.

Peaktew

Jan 2025 – April 2025

AI Software Developer

Waterloo, Canada

- Trained and evaluated **CNN and Transformer** models on 50K+ samples using **PyTorch/TensorFlow**, improving **validation accuracy by 8%** through structured data augmentation and hyperparameter tuning.
- Worked with **OpenAI, Hugging Face, and PyCryptodome** libraries to test model integration, verify encryption modules, and debug compatibility issues between inference APIs and local runtimes.
- Wrote unit tests for ML workflows and **mock inference scripts**, achieving **90% test coverage** across pipelines and ensuring consistent predictions during weekly model iterations.

Muide

May 2023 – Aug 2023

Full-Stack Developer

Waterloo, Canada

- Built analytics workflows in **Python and SQL** to monitor mentor-mentee engagement across 1,000+ sessions, identifying drop-off patterns and improving user retention by 12%.
- Automated weekly **A/B tests** and data reports in Looker Studio, **reducing manual analysis time by 30%** and enabling data-driven improvements to the matching algorithm.

Projects

MultiPoker | *Swift, Xcode, iOS, Game Logic, OOP*

- Built a modular poker engine supporting multiple game variants through reusable gameplay, player-action, and hand-evaluation logic, with clear separation between core systems and interface layers.

Agent Orchestrator – Distributed AI Execution Engine | *Go, SQLite, Claude APIs, Concurrency, REST*

- Engineered a distributed AI execution system for coordinating multi-step agent workflows with persistent state, retry handling, and modular workflow control for reliable automated reasoning pipelines.

AURA – Automated Understanding & Remediation for Audits | *Python, Rust, FastAPI, Slither, Mythril*

- Built an AI-assisted audit platform that analyzes codebases for vulnerabilities, synthesizes multi-tool findings, explains root causes, and generates remediation guidance to accelerate secure code review.

Technical Skills

Languages: Python, C#, Swift, Kotlin, JavaScript/TypeScript, SQL, Go, Rust, C++

AI/ML: Azure OpenAI, Semantic Kernel, Agentic AI, RAG, Embeddings, Prompt Engineering, LLM Evaluation, PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face

Backend/Cloud: .NET 8, Azure Functions, REST APIs, FastAPI, Docker, CI/CD, Serverless

Data/Storage: NumPy, Pandas, Cosmos DB, Blob Storage, PostgreSQL, MongoDB, Vector Databases